

Jinsuk Mo



Jinsuk Mo

Korea University (Sungbuk-gu, Seoul, Republic of Korea)
Tel) +82) 10-3063-9288
E-mail) mo9288@korea.ac.kr

EDUCATION

Department of Materials Science and Engineering, Korea University *Mar. 2023 – Present*
(4-semester hiatus due to mandatory military service)
Bachelor Student
Expected graduation: Feb. 2029
Culminative GPA: 4.42/4.5, Major GPA: 4.28/4.5

CONFERENCE PARTICIPATION (*: Contributed equally, †: Corresponding author)

-
- Suh, J.*, Lee, G.*, **Mo, J.***, Kim, G.*, Chun, Y.*, Kim, H.*, Kim, S.*, and Lee, M†, “An innovative strategy for the synthesis of single-atom catalysts on MXene surfaces”, 2024 *Fall Meeting of the Korean Ceramic Society* (17/10/2024)
 - Abstract book p. 307., PG3-8, Energy Environmental Ceramics

HONORS AND AWARDS

-
- Honorable Mention** *Mar. 2025 – Sep. 2025*
The 6th Corrosion Case Poster Contest 2025
The Corrosion Science Society of Korea
 - Research Topic:** Formation of Goethite layer for preventing rebar corrosion in marine atmosphere
 - Co-recipient of the 1st Place Award, Excellence Award** *Aug. 2024 – Oct. 2024*
Smart Convergence Materials Science and Technology Competition
Gwangju Institute of Science and Technology (GIST)
 - Research Topic:** Pd single-atom-coordinated g-C₃N₄ self-assembled partially oxidized Ti₃C₂ MXene type II-II heterojunction photocatalyst for reduction and conversion of greenhouse gas
 - Team Leader, Presenter
 - Advisor:** Se-Ho Kim (Department of Materials Science and Engineering, Korea University)
 - 3rd Place Award, Excellence Award** *Aug. 2024*
2nd Carbon Neutral Ideathon with Korea South-East Power Co. Ltd.
Korea South-East Power Co. Ltd. With participant universities in the Leaders in Industry-University Cooperation 3.0 (LINC 3.0): **Korea University, Kangwon National University, and Pohang Institute of Science and Technology (POSTECH)**
 - Research Topic:** Facility for microalgae-based energy production utilizing unused land after the modernization of the power plant
 - Team Leader, Presenter
 - Advisor:** Sanghun Sul (Applied Innovation Experience Design (AIEX), Sungkyunkwan University)
 - Dean's List** *Fall 2023*
 - Semester High Honors** *Fall 2024, Spring 2024, Fall 2023, Spring 2023*

Ongoing Project

-
- Campus Patent Universiade *Apr. 2026 -*
Korea Invention Promotion Association
 - Research Topic:** Patent Landscape and Technology Roadmap of HfO₂-based Ferroelectric memory devices
 - Investigated a patent landscape and technology roadmap with a focus on 3D vertical architectures

- Proposed technology strategies including interface reduction, oxygen-vacancy defect engineering, and future candidates of MFS-compatible channel materials
- **Advisor:** Min-kyu Song (Department of Electrical Engineering, Korea University)

SCHOLARSHIPS

- **SPECIAL SCHOLARSHIPS** *Fall 2024, Fall 2023*
Ministry of Student Affairs, Korea University
 - Recognized for past achievements and future potential
- **BEST HONORS SCHOLARSHIPS** *Spring 2024*
College of Engineering, Korea University
 - Recognized for academic excellence and outstanding performance
- **Daesung Mimir Scholarships** *Spring 2023*
Digital Daesung
 - Scholarship for students with superior College Scholastic Ability Test (CSAT) score

MAJOR RELATED EXPERIENCES

- **Kinetic Investigations via Cyclic Voltammetry** *Dec. 2024 – Jan. 2025*
 - **Research Topic:** Quinone/Hydroquinone redox reactions on a carbon paper electrode in sulfuric acid electrolyte
 - Analyzed hydrophilic surface functionalization of carbon paper via cyclic voltammetry
 - **Instructor:** Woong Kim (Department of Materials Science and Engineering, Korea University)
- **SK Hynix Campus Meet Up Day** *Aug. 2024*
SK Hynix
 - Participant in a corporate training program of Processor Engineer role at SK Hynix
 - <https://www.skcareersjournal.com/m/3328>
- **Alchemist Future Innovation Idea Competition** *Feb. 2024 – May. 2024*
Ministry of Trade, Industry and Energy and Korea Planning & Evaluation Institute of Industrial Technology (KEIT)
 - Government-funded project aimed at developing foundational technologies
 - **Research Topic:** Reversible meta-surfaces and IR laser for optical communication
 - Team Member, Presenter
 - **Advisor:** Yongnam Song (Department of Mechanical Engineering, Korea University)
- **Samsung Shining Star** *Jan. 2024*
Samsung Electronics DS
 - Participant in a corporate training program

EXTRACURRICULAR ACTIVITIES

- **Korea University AA CO-E Creator Club** *Jul. 2024 – Present*
 - 1st Semester: Acquired basic knowledge of Python and VBA
Worked on a team project based on a random number generation algorithm
- **Transistor Education and Hands-on Training** *Feb. 2025*
Korea Semiconductor Industry Association (KISA)
 - Completed training course on semiconductor devices and processes, including transistor circuit design and electrical characteristic analysis (for 30 hours, 5 days)
- **Understanding of the Power Generation** *Aug. 2024*
Korea Institute of Power Human Resource Development (KIPHRD)
 - Completed training course on power plant, including gas turbine system, HRSG system, renewable energy, and energy transport (for 10 hours, online)
- **Nanoscale Fabrication and Manufacturing, K-MOOC** *Aug. 2024*
Pohang Institute of Science and Technology (POSTECH) and K-MOOC
 - Acquired advanced knowledge of semiconductor processing (Score: 86.67/100)
- **Big Data Training and Practice** *Jul. 2024*
Innovation Center for Engineering Education, Korea University
 - Completed training course on rudimentary data analysis using python (for 40 hours, 5 days)
- **Introduction to Secondary Battery** *Jul. 2024*
Innovation Center for Engineering Education, Korea University
 - Completed training course on electrochemistry for battery, including electrode materials and

electrolyte materials (for 40 hours, 5 days)

Reciprocal Teaching

Mar. 2024 – Jun. 2024

- Studied a course *Introduction to Ceramics* collaboratively

SKILLS AND TECHNIQUES

Experimental Techniques

- **Electrochemistry:** cyclic voltammetry
- **Finite Difference Method:** coursework for course *Kinetics for Materials Science*

Certificates

- TOEFL 110/120 (ETS, 2026)
- Goethe Zertifikat B1 (Goethe Institut, 2026)
- Computer Specialist in Spreadsheet and Database (Korea Chamber of Commerce and Industry, 2025)
- Advanced Data analysis Semi-Professional (K-Data, 2024)
- Driver's license (2022)

Languages

- Korean (Native), English (CEFR C1), German (CEFR B1)